computer vision ex 4.

1a) The line function that the 2 pixels created is:

$$\begin{cases} (x_0, y_0) = (-2, 1) \Leftrightarrow y = -a x_0 + y_0 = 2a + 1 \\ (x_1, y_1) = (2, 5) \Leftrightarrow y = -a x_1 + y_1 = -2a + 5 \end{cases}$$

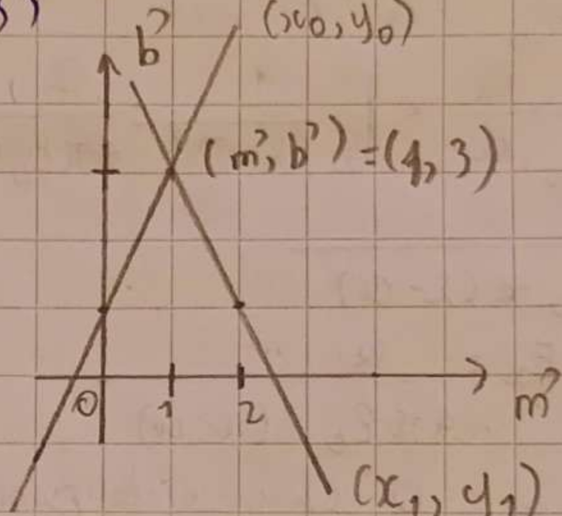
The point of intersection is:

$$2a + 1 = -2a + 5 \Rightarrow a = 1 \Rightarrow m' = 1$$

we have: $b' = y \Rightarrow b' = 2 \times 1 + 1 = 3$

$\Rightarrow (m', b') = (1, 3)$

plot:



1b) In polar coordinate parameter space, a line's form is: $p = x\cos\theta + y\sin\theta$

$\Rightarrow$ The sinusoids the points create has the forms:

$$\begin{cases} (x_0, y_0) = (-2, 1) \Leftrightarrow p = -2\cos\theta + \sin\theta \\ (x_1, y_1) = (2, 5) \Leftrightarrow p = 2\cos\theta + 5\sin\theta \end{cases}$$

The intersection point when $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$ is:

$$-2\cos\theta + \sin\theta = 2\cos\theta + 5\sin\theta$$

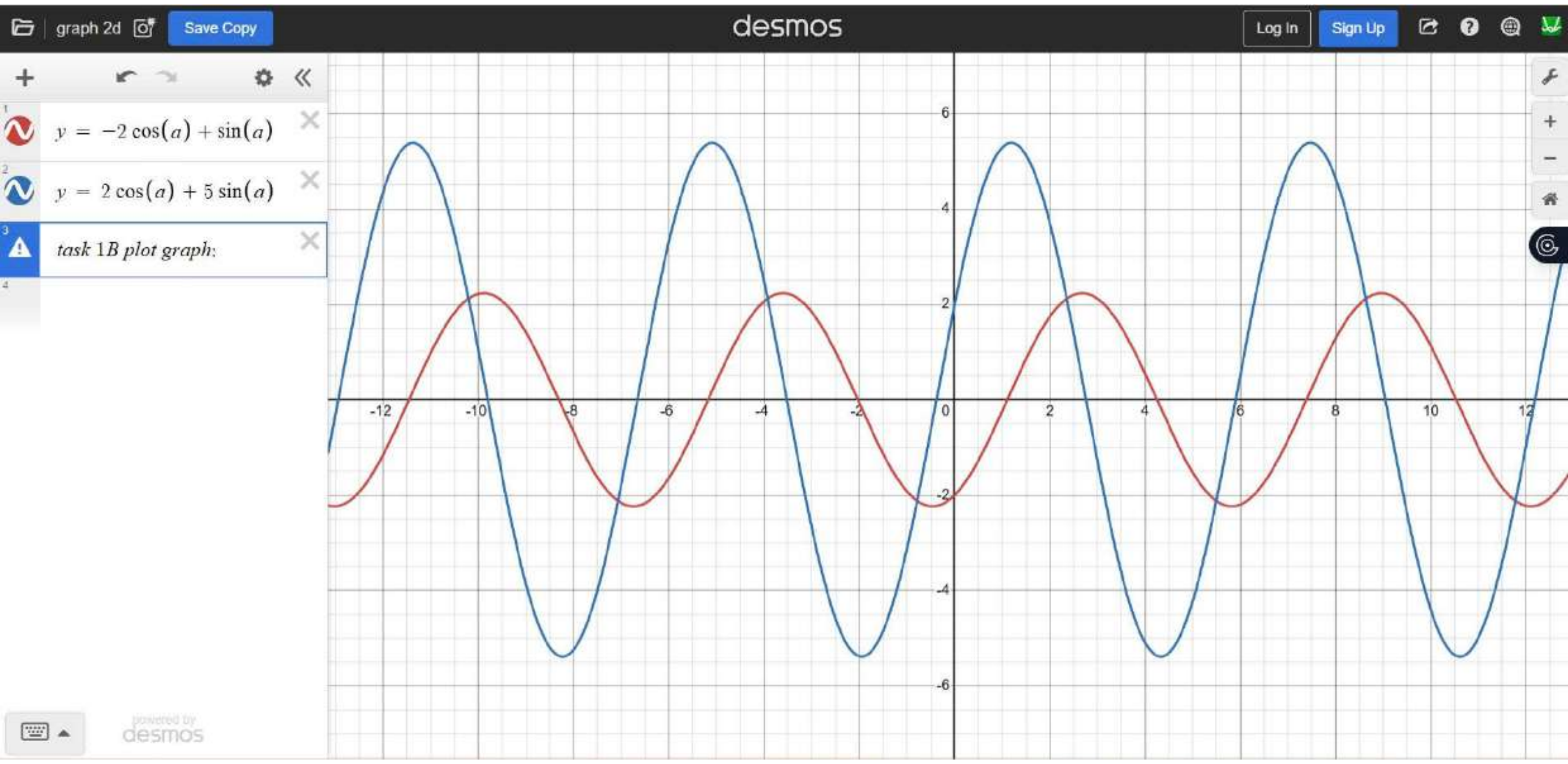
$$\Rightarrow -4\cos\theta = 4\sin\theta \Rightarrow \sin\theta / \cos\theta = -1$$

$$\Rightarrow \tan\theta = -1 \Rightarrow \theta = \arctan(-1) = -\frac{\pi}{4}$$

$$\Rightarrow \theta' = -\frac{\pi}{4} \Rightarrow p' = 2\cos\left(-\frac{\pi}{4}\right) + 5\sin\left(-\frac{\pi}{4}\right) = -\frac{3\sqrt{2}}{2}$$

$\Rightarrow$ The intersection point is: $(\theta', p') = \left(-\frac{\pi}{4}, \frac{-3\sqrt{2}}{2}\right)$

1c) In polar coordinate space, the lines are curves and therefore they don't ~~be com~~ have infinite slope $\Rightarrow$ Computers store information easier and more stable


graph 2d
Save Copy
desmos
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$y = -2\cos(a) + \sin(a)$
$y = 2\cos(a) + 5\sin(a)$
task 1B plot graph:
powered by desmos
6
4
2
-2
-4
-6
-12
-10
-8
-6
-4
-2
0
2
4
6
8
10
12